UM10360

Chapter 7: LPC176x/5x Pin configuration

7.1 LPC176x/5x pin configuration

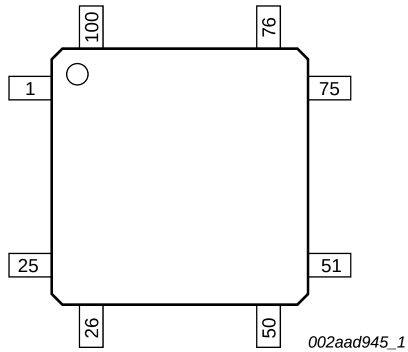

Fig 14. LPC176x LQFP100 pin configuration

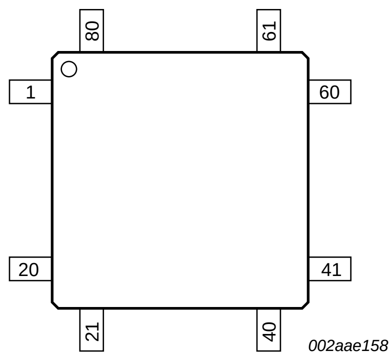

Fig 15. LPC175x LQFP80 pin configuration

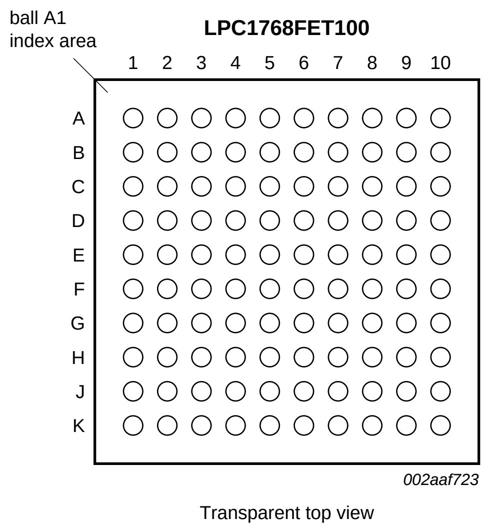

Fig 16. Pin configuration TFBGA100 package

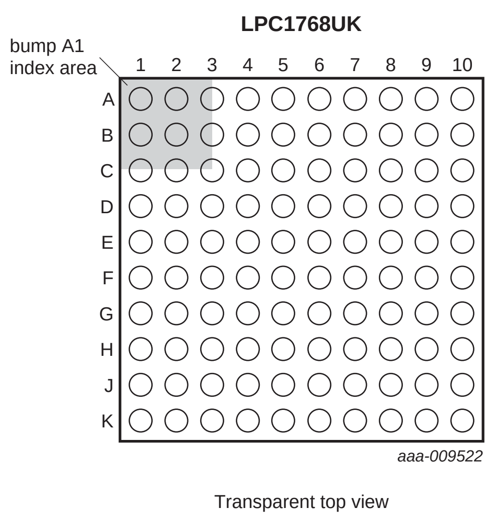

Fig 17. Pin configuration WLCSP100 package

Table 72. Pin allocation table TFBGA100 package

Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol
Row A							
1	TDO/SWO	2	P0[3]/RXD0/AD0[6]	3	$V_{DD(3V3)}$	4	P1[4]/ENET_TX_EN
5	P1[10]/ENET_RXD1	6	P1[16]/ENET_MDC	7	$V_{DD(REG)(3V3)}$	8	P0[4]/I2SRX_CLK/ RD2/CAP2[0]
9	P0[7]/I2STX_CLK/ SCK1/MAT2[1]	10	P0[9]/I2STX_SDA/ MOSI1/MAT2[3]	11	-	12	-
Row B							
1	TMS/SWDIO	2	RTCK	3	V_{SS}	4	P1[1]/ENET_TXD1

Table 72. Pin allocation table TFBGA100 package *…continued*

Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol
5	P1[9]/ENET_RXD0	6	P1[17]/ ENET_MDIO	7	V_{SS}	8	P0[6]/I2SRX_SDA/ SSEL1/MAT2[0]
9	P2[0]/PWM1[1]/TXD1	10	P2[1]/PWM1[2]/RXD1	11	-	12	-
Row C							
1	TCK/SWDCLK	2	$\overline{\text{TRST}}$	3	TDI	4	P0[2]/TXD0/AD0[7]
5	P1[8]/ENET_CRS	6	P1[15]/ ENET_REF_CLK	7	P4[28]/RX_MCLK/ MAT2[0]/TXD3	8	P0[8]/I2STX_WS/ MISO1/MAT2[2]
9	V_{SS}	10	$V_{DD(3V3)}$	11	-	12	-
Row D							
1	P0[24]/AD0[1]/ I2SRX_WS/CAP3[1]	2	P0[25]/AD0[2]/ I2SRX_SDA/TXD3	3	P0[26]/AD0[3]/ AOUT/RXD3	4	n.c.
5	P1[0]/ENET_TXD0	6	P1[14]/ENET_RX_ER	7	P0[5]/I2SRX_WS/ TD2/CAP2[1]	8	P2[2]/PWM1[3]/ CTS1/TRACEDATA[3]
9	P2[4]/PWM1[5]/ DSR1/TRACEDATA[1]	10	P2[5]/PWM1[6]/ DTR1/TRACEDATA[0]	11	-	12	-
Row E							
1	V_{SSA}	2	V_{DDA}	3	VREFP	4	n.c.
5	P0[23]/AD0[0]/ I2SRX_CLK/CAP3[0]	6	P4[29]/TX_MCLK/ MAT2[1]/RXD3	7	P2[3]/PWM1[4]/ DCD1/TRACEDATA[2]	8	P2[6]/PCAP1[0]/ RI1/TRACECLK
9	P2[7]/RD2/RTS1	10	P2[8]/TD2/TXD2	11	-	12	-
Row F							
1	VREFN	2	RTCX1	3	$\overline{\text{RESET}}$	4	P1[31]/SCK1/ AD0[5]
5	P1[21]/$\overline{\text{MCABORT}}$/ PWM1[3]/SSEL0	6	P0[18]/DCD1/ MOSI0/MOSI	7	P2[9]/USB_CONNECT/ RXD2	8	P0[16]/RXD1/ SSEL0/SSEL
9	P0[17]/CTS1/ MISO0/MISO	10	P0[15]/TXD1/ SCK0/SCK	11	-	12	-
Row G							
1	RTCX2	2	VBAT	3	XTAL2	4	P0[30]/USB_D−
5	P1[25]/MCOA1/ MAT1[1]	6	P1[29]/MCOB2/ PCAP1[1]/MAT0[1]	7	V_{SS}	8	P0[21]/RI1/RD1
9	P0[20]/DTR1/SCL1	10	P0[19]/DSR1/SDA1	11	-	12	-
Row H							
1	P1[30]/V_{BUS}/ AD0[4]	2	XTAL1	3	P3[25]/MAT0[0]/ PWM1[2]	4	P1[18]/USB_UP_LED/ PWM1[1]/CAP1[0]
5	P1[24]/MCI2/ PWM1[5]/MOSI0	6	$V_{DD(REG)(3V3)}$	7	P0[10]/TXD2/ SDA2/MAT3[0]	8	P2[11]/$\overline{\text{EINT1}}$/ I2STX_CLK
9	$V_{DD(3V3)}$	10	P0[22]/RTS1/TD1	11	-	12	-

Table 72. Pin allocation table TFBGA100 package *…continued*

Pin	Symbol	Pin	Symbol	Pin	Symbol	Pin	Symbol
Row J							
1	P0[28]/SCL0/ USB_SCL	2	P0[27]/SDA0/ USB_SDA	3	P0[29]/USB_D+	4	P1[19]/MCOA0/ USB_PPWR/ CAP1[1]
5	P1[22]/MCOB0/ USB_PWRD/ MAT1[0]	6	V_{SS}	7	P1[28]/MCOA2/ PCAP1[0]/ MAT0[0]	8	P0[1]/TD1/RXD3/SCL1
9	P2[13]/EINT3/ I2STX_SDA	10	P2[10]/EINT0/NMI	11	-	12	-
Row K							
1	P3[26]/STCLK/ MAT0[1]/PWM1[3]	2	$V_{DD(3V3)}$	3	V_{SS}	4	P1[20]/MCI0/ PWM1[2]/SCK0
5	P1[23]/MCI1/ PWM1[4]/MISO0	6	P1[26]/MCOB1/ PWM1[6]/CAP0[0]	7	P1[27]/CLKOUT /USB_OVRCR/ CAP0[1]	8	P0[0]/RD1/TXD3/SDA1
9	P0[11]/RXD2/ SCL2/MAT3[1]	10	P2[12]/EINT2/ I2STX_WS	11	-	12	-

7.1.1 LPC176x/5x pin description

I/O pins on the LPC176x/5x are 5V tolerant and have input hysteresis unless indicated in the table below. Crystal pins, power pins, and reference voltage pins are not 5V tolerant. In addition, when pins are selected to be A to D converter inputs, they are no longer 5V tolerant and must be limited to the voltage at the ADC positive reference pin (V_{REFP}).

Table 73. Pin description (LPC175x)

Symbol	Pin	Type	Description
P0[0] to P0[31]		I/O	**Port 0:** Port 0 is a 32-bit I/O port with individual direction controls for each bit. The operation of Port 0 pins depends upon the pin function selected via the pin connect block. Some port pins are not available on the LQFP80 package.
P0[0]/RD1/TXD3/ SDA1	37[1]	I/O	**P0[0] —** General purpose digital input/output pin.
		I	**RD1 —** CAN1 receiver input.
		O	**TXD3 —** Transmitter output for UART3.
		I/O	**SDA1 —** I^2C1 data input/output (this is not an I^2C-bus compliant open-drain pin).
P0[1]/TD1/RXD3/ SCL1	38[1]	I/O	**P0[1] —** General purpose digital input/output pin.
		O	**TD1 —** CAN1 transmitter output.
		I	**RXD3 —** Receiver input for UART3.
		I/O	**SCL1 —** I^2C1 clock input/output (this is not an I^2C-bus compliant open-drain pin).
P0[2]/TXD0/AD0[7]	79[2]	I/O	**P0[2] —** General purpose digital input/output pin.
		O	**TXD0 —** Transmitter output for UART0.
		I	**AD0[7] —** A/D converter 0, input 7.
P0[3]/RXD0/AD0[6]	80[2]	I/O	**P0[3] —** General purpose digital input/output pin.
		I	**RXD0 —** Receiver input for UART0.
		I	**AD0[6] —** A/D converter 0, input 6.

Table 73. Pin description (LPC175x) *…continued*

Symbol	Pin	Type	Description
P0[6]/ I2SRX_SDA/ SSEL1/MAT2[0]	64[1]	I/O	**P0[6] —** General purpose digital input/output pin.
		I/O	**I2SRX_SDA —** Receive data. It is driven by the transmitter and read by the receiver. Corresponds to the signal SD in the *I²S-bus specification*. (LPC1759/58/56 only).
		I/O	**SSEL1 —** Slave Select for SSP1.
		O	**MAT2[0] —** Match output for Timer 2, channel 0.
P0[7]/I2STX_CLK/ SCK1/MAT2[1]	63[1]	I/O	**P0[7] —** General purpose digital input/output pin.
		I/O	**I2STX_CLK —** Transmit Clock. It is driven by the master and received by the slave. Corresponds to the signal SCK in the *I²S-bus specification*. (LPC1759/58/56 only).
		I/O	**SCK1 —** Serial Clock for SSP1.
		O	**MAT2[1] —** Match output for Timer 2, channel 1.
P0[8]/I2STX_WS/ MISO1/MAT2[2]	62[1]	I/O	**P0[8] —** General purpose digital input/output pin.
		I/O	**I2STX_WS —** Transmit Word Select. It is driven by the master and received by the slave. Corresponds to the signal WS in the *I²S-bus specification*. (LPC1759/58/56 only).
		I/O	**MISO1 —** Master In Slave Out for SSP1.
		O	**MAT2[2] —** Match output for Timer 2, channel 2.
P0[9]/I2STX_SDA/ MOSI1/MAT2[3]	61[1]	I/O	**P0[9] —** General purpose digital input/output pin.
		I/O	**I2STX_SDA —** Transmit data. It is driven by the transmitter and read by the receiver. Corresponds to the signal SD in the *I²S-bus specification*. (LPC1759/58/56 only).
		I/O	**MOSI1 —** Master Out Slave In for SSP1.
		O	**MAT2[3] —** Match output for Timer 2, channel 3.
P0[10]/TXD2/ SDA2/MAT3[0]	39[1]	I/O	**P0[10] —** General purpose digital input/output pin.
		O	**TXD2 —** Transmitter output for UART2.
		I/O	**SDA2 —** I²C2 data input/output (this is not an open-drain pin).
		O	**MAT3[0] —** Match output for Timer 3, channel 0.
P0[11]/RXD2/ SCL2/MAT3[1]	40[1]	I/O	**P0[11] —** General purpose digital input/output pin.
		I	**RXD2 —** Receiver input for UART2.
		I/O	**SCL2 —** I²C2 clock input/output (this is not an open-drain pin).
		O	**MAT3[1] —** Match output for Timer 3, channel 1.
P0[15]/TXD1/ SCK0/SCK	47[1]	I/O	**P0[15] —** General purpose digital input/output pin.
		O	**TXD1 —** Transmitter output for UART1.
		I/O	**SCK0 —** Serial clock for SSP0.
		I/O	**SCK —** Serial clock for SPI.
P0[16]/RXD1/ SSEL0/SSEL	48[1]	I/O	**P0[16] —** General purpose digital input/output pin.
		I	**RXD1 —** Receiver input for UART1.
		I/O	**SSEL0 —** Slave Select for SSP0.
		I/O	**SSEL —** Slave Select for SPI.

Table 73. Pin description (LPC175x) *…continued*

Symbol	Pin	Type	Description
P0[17]/CTS1/ MISO0/MISO	46[1]	I/O	**P0[17] —** General purpose digital input/output pin.
		I	**CTS1 —** Clear to Send input for UART1.
		I/O	**MISO0 —** Master In Slave Out for SSP0.
		I/O	**MISO —** Master In Slave Out for SPI.
P0[18]/DCD1/ MOSI0/MOSI	45[1]	I/O	**P0[18] —** General purpose digital input/output pin.
		I	**DCD1 —** Data Carrier Detect input for UART1.
		I/O	**MOSI0 —** Master Out Slave In for SSP0.
		I/O	**MOSI —** Master Out Slave In for SPI.
P0[22]/RTS1/TD1	44[1]	I/O	**P0[22] —** General purpose digital input/output pin.
		O	**RTS1 —** Request to Send output for UART1. Can also be configured to be an RS-485/EIA-485 output enable signal.
		O	**TD1 —** CAN1 transmitter output.
P0[25]/AD0[2]/ I2SRX _SDA/ TXD3	7[2]	I/O	**P0[25] —** General purpose digital input/output pin.
		I	**AD0[2] —** A/D converter 0, input 2.
		I/O	**I2SRX_SDA —** Receive data. It is driven by the transmitter and read by the receiver. Corresponds to the signal SD in the *I^2S-bus specification*. (LPC1759/58/56 only).
		O	**TXD3 —** Transmitter output for UART3.
P0[26]/AD0[3]/ AOUT/RXD3	6[3]	I/O	**P0[26] —** General purpose digital input/output pin.
		I	**AD0[3] —** A/D converter 0, input 3.
		O	**AOUT —** DAC output. (LPC1759/58/56/54 only).
		I	**RXD3 —** Receiver input for UART3.
P0[29]/USB_D+	22[4]	I/O	**P0[29] —** General purpose digital input/output pin.
		I/O	**USB_D+ —** USB bidirectional D+ line.
P0[30]/USB_D−	23[4]	I/O	**P0[30] —** General purpose digital input/output pin.
		I/O	**USB_D− —** USB bidirectional D− line.
P1[0] to P1[31]		I/O	**Port 1:** Port 1 is a 32-bit I/O port with individual direction controls for each bit. The operation of port 1 pins depends upon the pin function selected via the pin connect block. Some port pins are not available on the LQFP80 package.
P1[0]/ ENET_TXD0	76[1]	I/O	**P1[0] —** General purpose digital input/output pin.
		O	**ENET_TXD0 —** Ethernet transmit data 0. (LPC1758 only).
P1[1]/ ENET_TXD1	75[1]	I/O	**P1[1] —** General purpose digital input/output pin.
		O	**ENET_TXD1 —** Ethernet transmit data 1. (LPC1758 only).
P1[4]/ ENET_TX_EN	74[1]	I/O	**P1[4] —** General purpose digital input/output pin.
		O	**ENET_TX_EN —** Ethernet transmit data enable. (LPC1758 only).
P1[8]/ ENET_CRS	73[1]	I/O	**P1[8] —** General purpose digital input/output pin.
		I	**ENET_CRS —** Ethernet carrier sense. (LPC1758 only).
P1[9]/ ENET_RXD0	72[1]	I/O	**P1[9] —** General purpose digital input/output pin.
		I	**ENET_RXD0 —** Ethernet receive data. (LPC1758 only).
P1[10]/ ENET_RXD1	71[1]	I/O	**P1[10] —** General purpose digital input/output pin.
		I	**ENET_RXD1 —** Ethernet receive data. (LPC1758 only).

Table 73. Pin description (LPC175x) *...continued*

Symbol	Pin	Type	Description
P1[14]/ ENET_RX_ER	70[1]	I/O	**P1[14] —** General purpose digital input/output pin.
		I	**ENET_RX_ER —** Ethernet receive error. (LPC1758 only).
P1[15]/ ENET_REF_CLK	69[1]	I/O	**P1[15] —** General purpose digital input/output pin.
		I	**ENET_REF_CLK —** Ethernet reference clock. (LPC1758 only).
P1[18]/ USB_UP_LED/ PWM1[1]/ CAP1[0]	25[1]	I/O	**P1[18] —** General purpose digital input/output pin.
		O	**USB_UP_LED —** USB GoodLink LED indicator. It is LOW when the device is configured (non-control endpoints enabled), or when the host is enabled and has detected a device on the bus. It is HIGH when the device is not configured, or when host is enabled and has not detected a device on the bus, or during global suspend. It transitions between LOW and HIGH (flashes) when the host is enabled and detects activity on the bus.
		O	**PWM1[1] —** Pulse Width Modulator 1, channel 1 output.
		I	**CAP1[0] —** Capture input for Timer 1, channel 0.
P1[19]/MCOA0/ USB_PPWR CAP1[1]	26[1]	I/O	**P1[19] —** General purpose digital input/output pin.
		O	**MCOA0 —** Motor control PWM channel 0, output A.
		O	**USB_PPWR —** Port Power enable signal for USB port. (LPC1759/58/56/54 only).
		I	**CAP1[1] —** Capture input for Timer 1, channel 1.
P1[20]/MCI0/ PWM1[2]/SCK0	27[1]	I/O	**P1[20] —** General purpose digital input/output pin.
		I	**MCI0 —** Motor control PWM channel 0, input. Also Quadrature Encoder Interface PHA input.
		O	**PWM1[2] —** Pulse Width Modulator 1, channel 2 output.
		I/O	**SCK0 —** Serial clock for SSP0.
P1[22]/MCOB0/ USB_PWRD/ MAT1[0]	28[1]	I/O	**P1[22] —** General purpose digital input/output pin.
		O	**MCOB0 —** Motor control PWM channel 0, output B.
		I	**USB_PWRD —** Power Status for USB port (host power switch). (LPC1759/58/56/54 only).
		O	**MAT1[0] —** Match output for Timer 1, channel 0.
P1[23]/MCI1/ PWM1[4]/MISO0	29[1]	I/O	**P1[23] —** General purpose digital input/output pin.
		I	**MCI1 —** Motor control PWM channel 1, input. Also Quadrature Encoder Interface PHB input.
		O	**PWM1[4] —** Pulse Width Modulator 1, channel 4 output.
		I/O	**MISO0 —** Master In Slave Out for SSP0.
P1[24]/MCI2/ PWM1[5]/MOSI0	30[1]	I/O	**P1[24] —** General purpose digital input/output pin.
		I	**MCI2 —** Motor control PWM channel 2, input. Also Quadrature Encoder Interface INDEX input.
		O	**PWM1[5] —** Pulse Width Modulator 1, channel 5 output.
		I/O	**MOSI0 —** Master Out Slave in for SSP0.
P1[25]/MCOA1/ MAT1[1]	31[1]	I/O	**P1[25] —** General purpose digital input/output pin.
		O	**MCOA1 —** Motor control PWM channel 1, output A.
		O	**MAT1[1] —** Match output for Timer 1, channel 1.

Table 73. Pin description (LPC175x) *…continued*

Symbol	Pin	Type	Description
P1[26]/MCOB1/ PWM1[6]/CAP0[0]	32[1]	I/O	**P1[26]** — General purpose digital input/output pin.
		O	**MCOB1** — Motor control PWM channel 1, output B.
		O	**PWM1[6]** — Pulse Width Modulator 1, channel 6 output.
		I	**CAP0[0]** — Capture input for Timer 0, channel 0.
P1[28]/MCOA2/ PCAP1[0]/ MAT0[0]	35[1]	I/O	**P1[28]** — General purpose digital input/output pin.
		O	**MCOA2** — Motor control PWM channel 2, output A.
		I	**PCAP1[0]** — Capture input for PWM1, channel 0.
		O	**MAT0[0]** — Match output for Timer 0, channel 0.
P1[29]/MCOB2/ PCAP1[1]/ MAT0[1]	36[1]	I/O	**P1[29]** — General purpose digital input/output pin.
		O	**MCOB2** — Motor control PWM channel 2, output B.
		I	**PCAP1[1]** — Capture input for PWM1, channel 1.
		O	**MAT0[1]** — Match output for Timer 0, channel 1.
P1[30]/V_{BUS}/ AD0[4]	18[2]	I/O	**P1[30]** — General purpose digital input/output pin.
		I	**V_{BUS}** — Monitors the presence of USB bus power. **Note:** This signal must be HIGH for USB reset to occur.
		I	**AD0[4]** — A/D converter 0, input 4.
P1[31]/SCK1/ AD0[5]	17[2]	I/O	**P1[31]** — General purpose digital input/output pin.
		I/O	**SCK1** — Serial Clock for SSP1.
		I	**AD0[5]** — A/D converter 0, input 5.
P2[0] to P2[31]		I/O	**Port 2:** Port 2 is a 32-bit I/O port with individual direction controls for each bit. The operation of port 2 pins depends upon the pin function selected via the pin connect block. Some port pins are not available on the LQFP80 package.
P2[0]/PWM1[1]/ TXD1	60[1]	I/O	**P2[0]** — General purpose digital input/output pin.
		O	**PWM1[1]** — Pulse Width Modulator 1, channel 1 output.
		O	**TXD1** — Transmitter output for UART1.
P2[1]/PWM1[2]/ RXD1	59[1]	I/O	**P2[1]** — General purpose digital input/output pin.
		O	**PWM1[2]** — Pulse Width Modulator 1, channel 2 output.
		I	**RXD1** — Receiver input for UART1.
P2[2]/PWM1[3]/ CTS1/ TRACEDATA[3]	58[1]	I/O	**P2[2]** — General purpose digital input/output pin.
		O	**PWM1[3]** — Pulse Width Modulator 1, channel 3 output.
		I	**CTS1** — Clear to Send input for UART1.
		O	**TRACEDATA[3]** — Trace data, bit 3.
P2[3]/PWM1[4]/ DCD1/ TRACEDATA[2]	55[1]	I/O	**P2[3]** — General purpose digital input/output pin.
		O	**PWM1[4]** — Pulse Width Modulator 1, channel 4 output.
		I	**DCD1** — Data Carrier Detect input for UART1.
		O	**TRACEDATA[2]** — Trace data, bit 2.
P2[4]/PWM1[5]/ DSR1/ TRACEDATA[1]	54[1]	I/O	**P2[4]** — General purpose digital input/output pin.
		O	**PWM1[5]** — Pulse Width Modulator 1, channel 5 output.
		I	**DSR1** — Data Set Ready input for UART1.
		O	**TRACEDATA[1]** — Trace data, bit 1.

Table 73. Pin description (LPC175x) *…continued*

Symbol	Pin	Type	Description
P2[5]/PWM1[6]/ DTR1/ TRACEDATA[0]	53[1]	I/O	**P2[5] —** General purpose digital input/output pin.
		O	**PWM1[6] —** Pulse Width Modulator 1, channel 6 output.
		O	**DTR1 —** Data Terminal Ready output for UART1. Can also be configured to be an RS-485/EIA-485 output enable signal.
		O	**TRACEDATA[0] —** Trace data, bit 0.
P2[6]/PCAP1[0]/ RI1/TRACECLK	52[1]	I/O	**P2[6] —** General purpose digital input/output pin.
		I	**PCAP1[0] —** Capture input for PWM1, channel 0.
		I	**RI1 —** Ring Indicator input for UART1.
		O	**TRACECLK —** Trace Clock.
P2[7]/RD2/ RTS1	51[1]	I/O	**P2[7] —** General purpose digital input/output pin.
		I	**RD2 —** CAN2 receiver input. (LPC1759/58/56 only).
		O	**RTS1 —** Request to Send output for UART1. Can also be configured to be an RS-485/EIA-485 output enable signal.
P2[8]/TD2/ TXD2	50[1]	I/O	**P2[8] —** General purpose digital input/output pin.
		O	**TD2 —** CAN2 transmitter output. (LPC1759/58/56 only).
		O	**TXD2 —** Transmitter output for UART2.
P2[9]/ USB_CONNECT/ RXD2	49[1]	I/O	**P2[9] —** General purpose digital input/output pin.
		O	**USB_CONNECT —** Signal used to switch an external 1.5 kΩ resistor under software control. Used with the SoftConnect USB feature.
		I	**RXD2 —** Receiver input for UART2.
P2[10]/$\overline{\text{EINT0}}$/NMI	41[5]	I/O	**P2[10] —** General purpose digital input/output pin. A LOW level on this pin during reset starts the ISP command handler.
		I	**$\overline{\text{EINT0}}$ —** External interrupt 0 input.
		I	**NMI —** Non-maskable interrupt input.
P4[0] to P4[31]		I/O	**Port 4:** Port 4 is a 32-bit I/O port with individual direction controls for each bit. The operation of port 4 pins depends upon the pin function selected via the pin connect block. Some port pins are not available on the LQFP80 package.
P4[28]/RX_MCLK/ MAT2[0]/TXD3	65[1]	I/O	**P4[28] —** General purpose digital input/output pin.
		O	**RX_MCLK —** I^{2}S receive master clock. (LPC1759/58/56 only).
		O	**MAT2[0] —** Match output for Timer 2, channel 0.
		O	**TXD3 —** Transmitter output for UART3.
P4[29]/TX_MCLK/ MAT2[1]/RXD3	68[1]	I/O	**P4[29] —** General purpose digital input/output pin.
		O	**TX_MCLK —** I^{2}S transmit master clock. (LPC1759/58/56 only).
		O	**MAT2[1] —** Match output for Timer 2, channel 1.
		I	**RXD3 —** Receiver input for UART3.
TDO/SWO	1[6]	O	**TDO —** Test Data out for JTAG interface.
		O	**SWO —** Serial wire trace output.
TDI	2[7]	I	**TDI —** Test Data in for JTAG interface.
TMS/SWDIO	3[7]	I	**TMS —** Test Mode Select for JTAG interface.
		I/O	**SWDIO —** Serial wire debug data input/output.
$\overline{\text{TRST}}$	4[7]	I	**$\overline{\text{TRST}}$ —** Test Reset for JTAG interface.

Table 73. Pin description (LPC175x) *…continued*

Symbol	Pin	Type	Description
TCK/SWDCLK	5[6]	I	**TCK —** Test Clock for JTAG interface.
		I	**SWDCLK —** Serial wire clock.
$\overline{\text{RSTOUT}}$	11	O	**$\overline{\text{RSTOUT}}$ —** This is a 3.3 V pin. LOW on this pin indicates UM10360 being in Reset state.
$\overline{\text{RESET}}$	14[8]	I	**External reset input:** A LOW-going pulse as short as 50 ns on this pin resets the device, causing I/O ports and peripherals to take on their default states, and processor execution to begin at address 0. TTL with hysteresis, 5 V tolerant.
XTAL1	19[9][10]	I	Input to the oscillator circuit and internal clock generator circuits.
XTAL2	20[9][10]	O	Output from the oscillator amplifier.
RTCX1	13[9][11]	I	Input to the RTC oscillator circuit.
RTCX2	15[9]	O	Output from the RTC oscillator circuit.
V_{SS}	24, 33, 43, 57, 66, 78	I	**ground:** 0 V reference.
V_{SSA}	9	I	**analog ground:** 0 V reference. This should nominally be the same voltage as V_{SS}, but should be isolated to minimize noise and error.
$V_{DD(3V3)}$	21, 42, 56, 77	I	**3.3 V supply voltage:** This is the power supply voltage for the I/O ports.
$V_{DD(REG)(3V3)}$	34, 67	I	**3.3 V voltage regulator supply voltage:** This is the supply voltage for the on-chip voltage regulator only.
V_{DDA}	8	I	**analog 3.3 V pad supply voltage:** This should be nominally the same voltage as $V_{DD(3V3)}$ but should be isolated to minimize noise and error. This voltage is used to power the ADC and DAC. This pin should be tied to 3.3 V if the ADC and DAC are not used.
VREFP	10	I	**ADC positive reference voltage:** This should be nominally the same voltage as V_{DDA} but should be isolated to minimize noise and error. Level on this pin is used as a reference for ADC and DAC. This pin should be tied to 3.3 V if the ADC and DAC are not used.
VREFN	12	I	**ADC negative reference voltage:** This should be nominally the same voltage as V_{SS} but should be isolated to minimize noise and error. Level on this pin is used as a reference for ADC and DAC.
VBAT	16[11]	I	**RTC pin power supply:** 3.3 V on this pin supplies the power to the RTC peripheral.

[1] 5 V tolerant pad providing digital I/O functions with TTL levels and hysteresis. This pin is pulled up to a voltage level of 2.3 V to 2.6 V.

[2] 5 V tolerant pad providing digital I/O functions (with TTL levels and hysteresis) and analog input. When configured as a ADC input, digital section of the pad is disabled and the pin is not 5 V tolerant. This pin is pulled up to a voltage level of 2.3 V to 2.6 V.

[3] 5 V tolerant pad providing digital I/O with TTL levels and hysteresis and analog output function. When configured as the DAC output, digital section of the pad is disabled. This pin is pulled up to a voltage level of 2.3 V to 2.6 V.

[4] Pad provides digital I/O and USB functions. It is designed in accordance with the *USB specification, revision 2.0* (Full-speed and Low-speed mode only). This pad is not 5 V tolerant.

[5] 5 V tolerant pad with 10 ns glitch filter providing digital I/O functions with TTL levels and hysteresis. This pin is pulled up to a voltage level of 2.3 V to 2.6 V.

[6] 5 V tolerant pad with TTL levels and hysteresis. Internal pull-up and pull-down resistors disabled.

[7] 5 V tolerant pad with TTL levels and hysteresis and internal pull-up resistor.

[8] 5 V tolerant pad with 20 ns glitch filter providing digital I/O function with TTL levels and hysteresis.

[9] Pad provides special analog functionality. 32 kHz crystal oscillator must be used with the RTC.

[10] When the system oscillator is not used, connect XTAL1 and XTAL2 as follows: XTAL1 can be left floating or can be grounded (grounding is preferred to reduce susceptibility to noise). XTAL2 should be left floating.

[11] When the RTC is not used, connect VBAT to $V_{DD(REG)(3V3)}$ and leave RTCX1 floating.

Table 74. Pin description (LPC176x)

Symbol	Pin/ball LQFP100	TFBGA100	WLCSP100		Type	Description
P0[0] to P0[31]					I/O	**Port 0:** Port 0 is a 32-bit I/O port with individual direction controls for each bit. The operation of port 0 pins depends upon the pin function selected via the pin connect block. Pins 12, 13, 14, and 31 of this port are not available.
P0[0]/RD1/TXD3/ SDA1	46	K8	H10	[1]	I/O	**P0[0]** — General purpose digital input/output pin.
					I	**RD1** — CAN1 receiver input. (LPC1769/68/66/65/64 only).
					O	**TXD3** — Transmitter output for UART3.
					I/O	**SDA1** — I²C1 data input/output. (This is not an I²C-bus compliant open-drain pin).
P0[1]/TD1/RXD3/ SCL1	47	J8	H9	[1]	I/O	**P0[1]** — General purpose digital input/output pin.
					O	**TD1** — CAN1 transmitter output. (LPC1769/68/66/65/64 only).
					I	**RXD3** — Receiver input for UART3.
					I/O	**SCL1** — I²C1 clock input/output. (This is not an I²C-bus compliant open-drain pin).
P0[2]/TXD0/AD0[7]	98	C4	B1	[2]	I/O	**P0[2]** — General purpose digital input/output pin.
					O	**TXD0** — Transmitter output for UART0.
					I	**AD0[7]** — A/D converter 0, input 7.
P0[3]/RXD0/AD0[6]	99	A2	C3	[2]	I/O	**P0[3]** — General purpose digital input/output pin.
					I	**RXD0** — Receiver input for UART0.
					I	**AD0[6]** — A/D converter 0, input 6.
P0[4]/ I2SRX_CLK/ RD2/CAP2[0]	81	A8	G2	[1]	I/O	**P0[4]** — General purpose digital input/output pin.
					I/O	**I2SRX_CLK** — Receive Clock. It is driven by the master and received by the slave. Corresponds to the signal SCK in the *I²S-bus specification*. (LPC1769/68/67/66/65/63 only).
					I	**RD2** — CAN2 receiver input. (LPC1769/68/66/65/64 only).
					I	**CAP2[0]** — Capture input for Timer 2, channel 0.
P0[5]/ I2SRX_WS/ TD2/CAP2[1]	80	D7	H1	[1]	I/O	**P0[5]** — General purpose digital input/output pin.
					I/O	**I2SRX_WS** — Receive Word Select. It is driven by the master and received by the slave. Corresponds to the signal WS in the *I²S-bus specification*. (LPC1769/68/67/66/65/63 only).
					O	**TD2** — CAN2 transmitter output. (LPC1769/68/66/65/64 only).
					I	**CAP2[1]** — Capture input for Timer 2, channel 1.

Table 74. Pin description (LPC176x) *...continued*

Symbol	LQFP100	TFBGA100	WLCSP100		Type	Description
P0[6]/ I2SRX_SDA/ SSEL1/MAT2[0]	79	B8	G3	[1]	I/O	**P0[6]** — General purpose digital input/output pin.
					I/O	**I2SRX_SDA** — Receive data. It is driven by the transmitter and read by the receiver. Corresponds to the signal SD in the *I²S-bus specification*. (LPC1769/68/67/66/65/63 only).
					I/O	**SSEL1** — Slave Select for SSP1.
					O	**MAT2[0]** — Match output for Timer 2, channel 0.
P0[7]/ I2STX_CLK/ SCK1/MAT2[1]	78	A9	J1	[1]	I/O	**P0[7]** — General purpose digital input/output pin.
					I/O	**I2STX_CLK** — Transmit Clock. It is driven by the master and received by the slave. Corresponds to the signal SCK in the *I²S-bus specification*. (LPC1769/68/67/66/65/63 only).
					I/O	**SCK1** — Serial Clock for SSP1.
					O	**MAT2[1]** — Match output for Timer 2, channel 1.
P0[8]/ I2STX_WS/ MISO1/MAT2[2]	77	C8	H2	[1]	I/O	**P0[8]** — General purpose digital input/output pin.
					I/O	**I2STX_WS** — Transmit Word Select. It is driven by the master and received by the slave. Corresponds to the signal WS in the *I²S-bus specification*. (LPC1769/68/67/66/65/63 only).
					I/O	**MISO1** — Master In Slave Out for SSP1.
					O	**MAT2[2]** — Match output for Timer 2, channel 2.
P0[9]/ I2STX_SDA/ MOSI1/MAT2[3]	76	A10	H3	[1]	I/O	**P0[9]** — General purpose digital input/output pin.
					I/O	**I2STX_SDA** — Transmit data. It is driven by the transmitter and read by the receiver. Corresponds to the signal SD in the *I²S-bus specification*. (LPC1769/68/67/66/65/63 only).
					I/O	**MOSI1** — Master Out Slave In for SSP1.
					O	**MAT2[3]** — Match output for Timer 2, channel 3.
P0[10]/TXD2/ SDA2/MAT3[0]	48	H7	H8	[1]	I/O	**P0[10]** — General purpose digital input/output pin.
					O	**TXD2** — Transmitter output for UART2.
					I/O	**SDA2** — I²C2 data input/output (this is not an open-drain pin).
					O	**MAT3[0]** — Match output for Timer 3, channel 0.
P0[11]/RXD2/ SCL2/MAT3[1]	49	K9	J10	[1]	I/O	**P0[11]** — General purpose digital input/output pin.
					I	**RXD2** — Receiver input for UART2.
					I/O	**SCL2** — I²C2 clock input/output (this is not an open-drain pin).
					O	**MAT3[1]** — Match output for Timer 3, channel 1.
P0[15]/TXD1/ SCK0/SCK	62	F10	H6	[1]	I/O	**P0[15]** — General purpose digital input/output pin.
					O	**TXD1** — Transmitter output for UART1.
					I/O	**SCK0** — Serial clock for SSP0.
					I/O	**SCK** — Serial clock for SPI.

Table 74. Pin description (LPC176x) *…continued*

Symbol	Pin/ball LQFP100	TFBGA100	WLCSP100		Type	Description
P0[16]/RXD1/ SSEL0/SSEL	63	F8	J5	[1]	I/O	**P0[16] —** General purpose digital input/output pin.
					I	**RXD1 —** Receiver input for UART1.
					I/O	**SSEL0 —** Slave Select for SSP0.
					I/O	**SSEL —** Slave Select for SPI.
P0[17]/CTS1/ MISO0/MISO	61	F9	K6	[1]	I/O	**P0[17] —** General purpose digital input/output pin.
					I	**CTS1 —** Clear to Send input for UART1.
					I/O	**MISO0 —** Master In Slave Out for SSP0.
					I/O	**MISO —** Master In Slave Out for SPI.
P0[18]/DCD1/ MOSI0/MOSI	60	F6	J6	[1]	I/O	**P0[18] —** General purpose digital input/output pin.
					I	**DCD1 —** Data Carrier Detect input for UART1.
					I/O	**MOSI0 —** Master Out Slave In for SSP0.
					I/O	**MOSI —** Master Out Slave In for SPI.
P0[19]/DSR1/ SDA1	59	G10	K7	[1]	I/O	**P0[19] —** General purpose digital input/output pin.
					I	**DSR1 —** Data Set Ready input for UART1.
					I/O	**SDA1 —** I^2C1 data input/output (this is not an I^2C-bus compliant open-drain pin).
P0[20]/DTR1/SCL1	58	G9	J7	[1]	I/O	**P0[20] —** General purpose digital input/output pin.
					O	**DTR1 —** Data Terminal Ready output for UART1. Can also be configured to be an RS-485/EIA-485 output enable signal.
					I/O	**SCL1 —** I^2C1 clock input/output (this is not an I^2C-bus compliant open-drain pin).
P0[21]/RI1/RD1	57	G8	H7	[1]	I/O	**P0[21] —** General purpose digital input/output pin.
					I	**RI1 —** Ring Indicator input for UART1.
					I	**RD1 —** CAN1 receiver input. (LPC1769/68/66/65/64 only).
P0[22]/RTS1/TD1	56	H10	K8	[1]	I/O	**P0[22] —** General purpose digital input/output pin.
					O	**RTS1 —** Request to Send output for UART1. Can also be configured to be an RS-485/EIA-485 output enable signal.
					O	**TD1 —** CAN1 transmitter output. (LPC1769/68/66/65/64 only).
P0[23]/AD0[0]/ I2SRX_CLK/ CAP3[0]	9	E5	D5	[2]	I/O	**P0[23] —** General purpose digital input/output pin.
					I	**AD0[0] —** A/D converter 0, input 0.
					I/O	**I2SRX_CLK —** Receive Clock. It is driven by the master and received by the slave. Corresponds to the signal SCK in the *I^2S-bus specification*. (LPC1769/68/67/66/65/63 only).
					I	**CAP3[0] —** Capture input for Timer 3, channel 0.

Table 74. Pin description (LPC176x) *…continued*

Symbol	Pin/ball LQFP100	TFBGA100	WLCSP100		Type	Description
P0[24]/AD0[1]/I2SRX_WS/CAP3[1]	8	D1	B4	[2]	I/O	**P0[24] —** General purpose digital input/output pin.
					I	**AD0[1] —** A/D converter 0, input 1.
					I/O	**I2SRX_WS —** Receive Word Select. It is driven by the master and received by the slave. Corresponds to the signal WS in the *I²S-bus specification*. (LPC1769/68/67/66/65/63 only).
					I	**CAP3[1] —** Capture input for Timer 3, channel 1.
P0[25]/AD0[2]/I2SRX_SDA/TXD3	7	D2	A3	[2]	I/O	**P0[25] —** General purpose digital input/output pin.
					I	**AD0[2] —** A/D converter 0, input 2.
					I/O	**I2SRX_SDA —** Receive data. It is driven by the transmitter and read by the receiver. Corresponds to the signal SD in the *I²S-bus specification*. (LPC1769/68/67/66/65/63 only).
					O	**TXD3 —** Transmitter output for UART3.
P0[26]/AD0[3]/AOUT/RXD3	6	D3	C5	[3]	I/O	**P0[26] —** General purpose digital input/output pin.
					I	**AD0[3] —** A/D converter 0, input 3.
					O	**AOUT —** DAC output (LPC1769/68/67/66/65/63 only).
					I	**RXD3 —** Receiver input for UART3.
P0[27]/SDA0/USB_SDA	25	J2	C8	[4]	I/O	**P0[27] —** General purpose digital input/output pin. Output is open-drain.
					I/O	**SDA0 —** I²C0 data input/output. Open-drain output (for I²C-bus compliance).
					I/O	**USB_SDA —** USB port I²C serial data (OTG transceiver, LPC1769/68/66/65 only).
P0[28]/SCL0/USB_SCL	24	J1	B9	[4]	I/O	**P0[28] —** General purpose digital input/output pin. Output is open-drain.
					I/O	**SCL0 —** I²C0 clock input/output. Open-drain output (for I²C-bus compliance).
					I/O	**USB_SCL —** USB port I²C serial clock (OTG transceiver, LPC1769/68/66/65 only).
P0[29]/USB_D+	29	J3	B10	[5]	I/O	**P0[29] —** General purpose digital input/output pin.
					I/O	**USB_D+ —** USB bidirectional D+ line. (LPC1769/68/66/65/64 only).
P0[30]/USB_D−	30	G4	C9	[5]	I/O	**P0[30] —** General purpose digital input/output pin.
					I/O	**USB_D− —** USB bidirectional D− line. (LPC1769/68/66/65/64 only).
P1[0] to P1[31]					I/O	**Port 1:** Port 1 is a 32-bit I/O port with individual direction controls for each bit. The operation of port 1 pins depends upon the pin function selected via the pin connect block. Pins 2, 3, 5, 6, 7, 11, 12, and 13 of this port are not available.
P1[0]/ENET_TXD0	95	D5	C1	[1]	I/O	**P1[0] —** General purpose digital input/output pin.
					O	**ENET_TXD0 —** Ethernet transmit data 0. (LPC1769/68/67/66/64 only).

Table 74. Pin description (LPC176x) *…continued*

Symbol	LQFP100	TFBGA100	WLCSP100		Type	Description
P1[1]/ ENET_TXD1	94	B4	C2	[1]	I/O	**P1[1]** — General purpose digital input/output pin.
					O	**ENET_TXD1** — Ethernet transmit data 1. (LPC1769/68/67/66/64 only).
P1[4]/ ENET_TX_EN	93	A4	D2	[1]	I/O	**P1[4]** — General purpose digital input/output pin.
					O	**ENET_TX_EN** — Ethernet transmit data enable. (LPC1769/68/67/66/64 only).
P1[8]/ ENET_CRS	92	C5	D1	[1]	I/O	**P1[8]** — General purpose digital input/output pin.
					I	**ENET_CRS** — Ethernet carrier sense. (LPC1769/68/67/66/64 only).
P1[9]/ ENET_RXD0	91	B5	D3	[1]	I/O	**P1[9]** — General purpose digital input/output pin.
					I	**ENET_RXD0** — Ethernet receive data. (LPC1769/68/67/66/64 only).
P1[10]/ ENET_RXD1	90	A5	E3	[1]	I/O	**P1[10]** — General purpose digital input/output pin.
					I	**ENET_RXD1** — Ethernet receive data. (LPC1769/68/67/66/64 only).
P1[14]/ ENET_RX_ER	89	D6	E2	[1]	I/O	**P1[14]** — General purpose digital input/output pin.
					I	**ENET_RX_ER** — Ethernet receive error. (LPC1769/68/67/66/64 only).
P1[15]/ ENET_REF_CLK	88	C6	E1	[1]	I/O	**P1[15]** — General purpose digital input/output pin.
					I	**ENET_REF_CLK** — Ethernet reference clock. (LPC1769/68/67/66/64 only).
P1[16]/ ENET_MDC	87	A6	F3	[1]	I/O	**P1[16]** — General purpose digital input/output pin.
					O	**ENET_MDC** — Ethernet MIIM clock (LPC1769/68/67/66/64 only).
P1[17]/ ENET_MDIO	86	B6	F2	[1]	I/O	**P1[17]** — General purpose digital input/output pin.
					I/O	**ENET_MDIO** — Ethernet MIIM data input and output. (LPC1769/68/67/66/64 only).
P1[18]/ USB_UP_LED/ PWM1[1]/ CAP1[0]	32	H4	D9	[1]	I/O	**P1[18]** — General purpose digital input/output pin.
					O	**USB_UP_LED** — USB GoodLink LED indicator. It is LOW when the device is configured (non-control endpoints enabled), or when the host is enabled and has detected a device on the bus. It is HIGH when the device is not configured, or when host is enabled and has not detected a device on the bus, or during global suspend. It transitions between LOW and HIGH (flashes) when the host is enabled and detects activity on the bus. (LPC1769/68/66/65/64 only).
					O	**PWM1[1]** — Pulse Width Modulator 1, channel 1 output.
					I	**CAP1[0]** — Capture input for Timer 1, channel 0.
P1[19]/MCOA0/ USB_PPWR/ CAP1[1]	33	J4	C10	[1]	I/O	**P1[19]** — General purpose digital input/output pin.
					O	**MCOA0** — Motor control PWM channel 0, output A.
					O	**USB_PPWR** — Port Power enable signal for USB port. (LPC1769/68/66/65 only).
					I	**CAP1[1]** — Capture input for Timer 1, channel 1.

Table 74. Pin description (LPC176x) *…continued*

Symbol	Pin/ball LQFP100	TFBGA100	WLCSP100		Type	Description
P1[20]/MCI0/PWM1[2]/SCK0	34	K4	E8	[1]	I/O	**P1[20]** — General purpose digital input/output pin.
					I	**MCI0** — Motor control PWM channel 0, input. Also Quadrature Encoder Interface PHA input.
					O	**PWM1[2]** — Pulse Width Modulator 1, channel 2 output.
					I/O	**SCK0** — Serial clock for SSP0.
P1[21]/MCABORT/PWM1[3]/SSEL0	35	F5	E9	[1]	I/O	**P1[21]** — General purpose digital input/output pin.
					O	**MCABORT** — Motor control PWM, LOW-active fast abort.
					O	**PWM1[3]** — Pulse Width Modulator 1, channel 3 output.
					I/O	**SSEL0** — Slave Select for SSP0.
P1[22]/MCOB0/USB_PWRD/MAT1[0]	36	J5	D10	[1]	I/O	**P1[22]** — General purpose digital input/output pin.
					O	**MCOB0** — Motor control PWM channel 0, output B.
					I	**USB_PWRD** — Power Status for USB port (host power switch, LPC1769/68/66/65 only).
					O	**MAT1[0]** — Match output for Timer 1, channel 0.
P1[23]/MCI1/PWM1[4]/MISO0	37	K5	E7	[1]	I/O	**P1[23]** — General purpose digital input/output pin.
					I	**MCI1** — Motor control PWM channel 1, input. Also Quadrature Encoder Interface PHB input.
					O	**PWM1[4]** — Pulse Width Modulator 1, channel 4 output.
					I/O	**MISO0** — Master In Slave Out for SSP0.
P1[24]/MCI2/PWM1[5]/MOSI0	38	H5	F8	[1]	I/O	**P1[24]** — General purpose digital input/output pin.
					I	**MCI2** — Motor control PWM channel 2, input. Also Quadrature Encoder Interface INDEX input.
					O	**PWM1[5]** — Pulse Width Modulator 1, channel 5 output.
					I/O	**MOSI0** — Master Out Slave in for SSP0.
P1[25]/MCOA1/MAT1[1]	39	G5	F9	[1]	I/O	**P1[25]** — General purpose digital input/output pin.
					O	**MCOA1** — Motor control PWM channel 1, output A.
					O	**MAT1[1]** — Match output for Timer 1, channel 1.
P1[26]/MCOB1/PWM1[6]/CAP0[0]	40	K6	E10	[1]	I/O	**P1[26]** — General purpose digital input/output pin.
					O	**MCOB1** — Motor control PWM channel 1, output B.
					O	**PWM1[6]** — Pulse Width Modulator 1, channel 6 output.
					I	**CAP0[0]** — Capture input for Timer 0, channel 0.
P1[27]/CLKOUT/USB_OVRCR/CAP0[1]	43	K7	G9	[1]	I/O	**P1[27]** — General purpose digital input/output pin.
					O	**CLKOUT** — Clock output pin.
					I	**USB_OVRCR** — USB port Over-Current status. (LPC1769/68/66/65 only).
					I	**CAP0[1]** — Capture input for Timer 0, channel 1.

Table 74. Pin description (LPC176x) *...continued*

Symbol	Pin/ball LQFP100	TFBGA100	WLCSP100		Type	Description
P1[28]/MCOA2/ PCAP1[0]/ MAT0[0]	44	J7	G10	[1]	I/O	**P1[28] —** General purpose digital input/output pin.
					O	**MCOA2 —** Motor control PWM channel 2, output A.
					I	**PCAP1[0] —** Capture input for PWM1, channel 0.
					O	**MAT0[0] —** Match output for Timer 0, channel 0.
P1[29]/MCOB2/ PCAP1[1]/ MAT0[1]	45	G6	G8	[1]	I/O	**P1[29] —** General purpose digital input/output pin.
					O	**MCOB2 —** Motor control PWM channel 2, output B.
					I	**PCAP1[1] —** Capture input for PWM1, channel 1.
					O	**MAT0[1] —** Match output for Timer 0, channel 1.
P1[30]/V_{BUS}/ AD0[4]	21	H1	B8	[2]	I/O	**P1[30] —** General purpose digital input/output pin.
					I	**V_{BUS} —** Monitors the presence of USB bus power. (LPC1769/68/66/65/64 only). **Note:** This signal must be HIGH for USB reset to occur.
					I	**AD0[4] —** A/D converter 0, input 4.
P1[31]/SCK1/ AD0[5]	20	F4	C7	[2]	I/O	**P1[31] —** General purpose digital input/output pin.
					I/O	**SCK1 —** Serial Clock for SSP1.
					I	**AD0[5] —** A/D converter 0, input 5.
P2[0] to P2[31]					I/O	**Port 2:** Port 2 is a 32-bit I/O port with individual direction controls for each bit. The operation of port 2 pins depends upon the pin function selected via the pin connect block. Pins 14 through 31 of this port are not available.
P2[0]/PWM1[1]/ TXD1	75	B9	K1	[1]	I/O	**P2[0] —** General purpose digital input/output pin.
					O	**PWM1[1] —** Pulse Width Modulator 1, channel 1 output.
					O	**TXD1 —** Transmitter output for UART1.
P2[1]/PWM1[2]/ RXD1	74	B10	J2	[1]	I/O	**P2[1] —** General purpose digital input/output pin.
					O	**PWM1[2] —** Pulse Width Modulator 1, channel 2 output.
					I	**RXD1 —** Receiver input for UART1.
P2[2]/PWM1[3]/ CTS1/ TRACEDATA[3]	73	D8	K2	[1]	I/O	**P2[2] —** General purpose digital input/output pin.
					O	**PWM1[3] —** Pulse Width Modulator 1, channel 3 output.
					I	**CTS1 —** Clear to Send input for UART1.
					O	**TRACEDATA[3] —** Trace data, bit 3.
P2[3]/PWM1[4]/ DCD1/ TRACEDATA[2]	70	E7	K3	[1]	I/O	**P2[3] —** General purpose digital input/output pin.
					O	**PWM1[4] —** Pulse Width Modulator 1, channel 4 output.
					I	**DCD1 —** Data Carrier Detect input for UART1.
					O	**TRACEDATA[2] —** Trace data, bit 2.
P2[4]/PWM1[5]/ DSR1/ TRACEDATA[1]	69	D9	J3	[1]	I/O	**P2[4] —** General purpose digital input/output pin.
					O	**PWM1[5] —** Pulse Width Modulator 1, channel 5 output.
					I	**DSR1 —** Data Set Ready input for UART1.
					O	**TRACEDATA[1] —** Trace data, bit 1.

Table 74. Pin description (LPC176x) *...continued*

Symbol	Pin/ball				Type	Description
	LQFP100	TFBGA100	WLCSP100			
P2[5]/PWM1[6]/ DTR1/ TRACEDATA[0]	68	D10	H4	[1]	I/O	**P2[5]** — General purpose digital input/output pin.
					O	**PWM1[6]** — Pulse Width Modulator 1, channel 6 output.
					O	**DTR1** — Data Terminal Ready output for UART1. Can also be configured to be an RS-485/EIA-485 output enable signal.
					O	**TRACEDATA[0]** — Trace data, bit 0.
P2[6]/PCAP1[0]/ RI1/TRACECLK	67	E8	K4	[1]	I/O	**P2[6]** — General purpose digital input/output pin.
					I	**PCAP1[0]** — Capture input for PWM1, channel 0.
					I	**RI1** — Ring Indicator input for UART1.
					O	**TRACECLK** — Trace Clock.
P2[7]/RD2/ RTS1	66	E9	J4	[1]	I/O	**P2[7]** — General purpose digital input/output pin.
					I	**RD2** — CAN2 receiver input. (LPC1769/68/66/65/64 only).
					O	**RTS1** — Request to Send output for UART1. Can also be configured to be an RS-485/EIA-485 output enable signal.
P2[8]/TD2/ TXD2	65	E10	H5	[1]	I/O	**P2[8]** — General purpose digital input/output pin.
					O	**TD2** — CAN2 transmitter output. (LPC1769/68/66/65/64 only).
					O	**TXD2** — Transmitter output for UART2.
P2[9]/ USB_CONNECT/ RXD2	64	F7	K5	[1]	I/O	**P2[9]** — General purpose digital input/output pin.
					O	**USB_CONNECT** — Signal used to switch an external 1.5 kΩ resistor under software control. Used with the SoftConnect USB feature. (LPC1769/68/66/65/64 only).
					I	**RXD2** — Receiver input for UART2.
P2[10]/$\overline{\text{EINT0}}$/NMI	53	J10	K9	[6]	I/O	**P2[10]** — General purpose digital input/output pin. A LOW level on this pin during reset starts the ISP command handler.
					I	**$\overline{\text{EINT0}}$** — External interrupt 0 input.
					I	**NMI** — Non-maskable interrupt input.
P2[11]/$\overline{\text{EINT1}}$/ I2STX_CLK	52	H8	J8	[6]	I/O	**P2[11]** — General purpose digital input/output pin.
					I	**$\overline{\text{EINT1}}$** — External interrupt 1 input.
					I/O	**I2STX_CLK** — Transmit Clock. It is driven by the master and received by the slave. Corresponds to the signal SCK in the *I²S-bus specification*. (LPC1769/68/67/66/65/63 only).
P2[12]/$\overline{\text{EINT2}}$/ I2STX_WS	51	K10	K10	[6]	I/O	**P2[12]** — General purpose digital input/output pin.
					I	**$\overline{\text{EINT2}}$** — External interrupt 2 input.
					I/O	**I2STX_WS** — Transmit Word Select. It is driven by the master and received by the slave. Corresponds to the signal WS in the *I²S-bus specification*. (LPC1769/68/67/66/65/63 only).
P2[13]/$\overline{\text{EINT3}}$/ I2STX_SDA	50	J9	J9	[6]	I/O	**P2[13]** — General purpose digital input/output pin.
					I	**$\overline{\text{EINT3}}$** — External interrupt 3 input.
					I/O	**I2STX_SDA** — Transmit data. It is driven by the transmitter and read by the receiver. Corresponds to the signal SD in the *I²S-bus specification*. (LPC1769/68/67/66/65/63 only).

Table 74. Pin description (LPC176x) *…continued*

Symbol	Pin/ball LQFP100	TFBGA100	WLCSP100		Type	Description
P3[0] to P3[31]					I/O	**Port 3:** Port 3 is a 32-bit I/O port with individual direction controls for each bit. The operation of port 3 pins depends upon the pin function selected via the pin connect block. Pins 0 through 24, and 27 through 31 of this port are not available.
P3[25]/MAT0[0]/PWM1[2]	27	H3	D8	[1]	I/O	**P3[25]** — General purpose digital input/output pin.
					O	**MAT0[0]** — Match output for Timer 0, channel 0.
					O	**PWM1[2]** — Pulse Width Modulator 1, output 2.
P3[26]/STCLK/MAT0[1]/PWM1[3]	26	K1	A10	[1]	I/O	**P3[26]** — General purpose digital input/output pin.
					I	**STCLK** — System tick timer clock input. The maximum STCLK frequency is 1/4 of the ARM processor clock frequency CCLK.
					O	**MAT0[1]** — Match output for Timer 0, channel 1.
					O	**PWM1[3]** — Pulse Width Modulator 1, output 3.
P4[0] to P4[31]					I/O	**Port 4:** Port 4 is a 32-bit I/O port with individual direction controls for each bit. The operation of port 4 pins depends upon the pin function selected via the pin connect block. Pins 0 through 27, 30, and 31 of this port are not available.
P4[28]/RX_MCLK/MAT2[0]/TXD3	82	C7	G1	[1]	I/O	**P4[28]** — General purpose digital input/output pin.
					O	**RX_MCLK** — I^2S receive master clock. (LPC1769/68/67/66/65 only).
					O	**MAT2[0]** — Match output for Timer 2, channel 0.
					O	**TXD3** — Transmitter output for UART3.
P4[29]/TX_MCLK/MAT2[1]/RXD3	85	E6	F1	[1]	I/O	**P4[29]** — General purpose digital input/output pin.
					O	**TX_MCLK** — I^2S transmit master clock. (LPC1769/68/67/66/65 only).
					O	**MAT2[1]** — Match output for Timer 2, channel 1.
					I	**RXD3** — Receiver input for UART3.
TDO/SWO	1	A1	A1	[1][7]	O	**TDO** — Test Data out for JTAG interface.
					O	**SWO** — Serial wire trace output.
TDI	2	C3	C4	[1][8]	I	**TDI** — Test Data in for JTAG interface.
TMS/SWDIO	3	B1	B3	[1][8]	I	**TMS** — Test Mode Select for JTAG interface.
					I/O	**SWDIO** — Serial wire debug data input/output.
$\overline{\text{TRST}}$	4	C2	A2	[1][8]	I	**$\overline{\text{TRST}}$** — Test Reset for JTAG interface.
TCK/SWDCLK	5	C1	D4	[1][7]	I	**TCK** — Test Clock for JTAG interface.
					I	**SWDCLK** — Serial wire clock.
RTCK	100	B2	B2	[1][7]	O	**RTCK** — JTAG interface control signal.
$\overline{\text{RSTOUT}}$	14	-	-	-	O	**$\overline{\text{RSTOUT}}$** — This is a 3.3 V pin. LOW on this pin indicates the microcontroller being in Reset state.
$\overline{\text{RESET}}$	17	F3	C6	[9]	I	**External reset input:** A LOW-going pulse as short as 50 ns on this pin resets the device, causing I/O ports and peripherals to take on their default states, and processor execution to begin at address 0. TTL with hysteresis, 5 V tolerant.

Table 74. Pin description (LPC176x) *…continued*

Symbol	Pin/ball LQFP100	TFBGA100	WLCSP100		Type	Description
XTAL1	22	H2	D7	[10][11]	I	Input to the oscillator circuit and internal clock generator circuits.
XTAL2	23	G3	A9	[10][11]	O	Output from the oscillator amplifier.
RTCX1	16	F2	A7	[10][11]	I	Input to the RTC oscillator circuit.
RTCX2	18	G1	B7	[10]	O	Output from the RTC oscillator circuit.
V_{SS}	31, 41, 55, 72, 83, 97	B3, B7, C9, G7, J6, K3	E5, F5, F6, G5, G6, G7	[10]	I	**ground:** 0 V reference.
V_{SSA}	11	E1	B5	[10]	I	**analog ground:** 0 V reference. This should nominally be the same voltage as V_{SS}, but should be isolated to minimize noise and error.
$V_{DD(3V3)}$	28, 54, 71, 96	K2, H9, C10 , A3	E4, E6, F7, G4	[10]	I	**3.3 V supply voltage:** This is the power supply voltage for the I/O ports.
$V_{DD(REG)(3V3)}$	42, 84	H6, A7	F4, F10	[10]	I	**3.3 V voltage regulator supply voltage:** This is the supply voltage for the on-chip voltage regulator only.
V_{DDA}	10	E2	A4	[10]	I	**analog 3.3 V pad supply voltage:** This should be nominally the same voltage as $V_{DD(3V3)}$ but should be isolated to minimize noise and error. This voltage is used to power the ADC and DAC. This pin should be tied to 3.3 V if the ADC and DAC are not used.
VREFP	12	E3	A5	[10]	I	**ADC positive reference voltage:** This should be nominally the same voltage as V_{DDA} but should be isolated to minimize noise and error. Level on this pin is used as a reference for ADC and DAC. This pin should be tied to 3.3 V if the ADC and DAC are not used.
VREFN	15	F1	A6		I	**ADC negative reference voltage:** This should be nominally the same voltage as V_{SS} but should be isolated to minimize noise and error. Level on this pin is used as a reference for ADC and DAC.
VBAT	19	G2	A8	[10]	I	**RTC pin power supply:** 3.3 V on this pin supplies the power to the RTC peripheral.
n.c.	13	D4, E4	B6, D6		-	not connected.

[1] 5 V tolerant pad providing digital I/O functions with TTL levels and hysteresis. This pin is pulled up to a voltage level of 2.3 V to 2.6 V.

[2] 5 V tolerant pad providing digital I/O functions (with TTL levels and hysteresis) and analog input. When configured as a ADC input, digital section of the pad is disabled and the pin is not 5 V tolerant. This pin is pulled up to a voltage level of 2.3 V to 2.6 V.

[3] 5 V tolerant pad providing digital I/O with TTL levels and hysteresis and analog output function. When configured as the DAC output, digital section of the pad is disabled. This pin is pulled up to a voltage level of 2.3 V to 2.6 V.

[4] Open-drain 5 V tolerant digital I/O pad, compatible with I^2C-bus 400 kHz specification. This pad requires an external pull-up to provide output functionality. When power is switched off, this pin connected to the I^2C-bus is floating and does not disturb the I^2C lines. Open-drain configuration applies to all functions on this pin.

[5] Pad provides digital I/O and USB functions. It is designed in accordance with the *USB specification, revision 2.0* (Full-speed and Low-speed mode only). This pad is not 5 V tolerant.

[6] 5 V tolerant pad with 10 ns glitch filter providing digital I/O functions with TTL levels and hysteresis. This pin is pulled up to a voltage level of 2.3 V to 2.6 V.

[7] 5 V tolerant pad with TTL levels and hysteresis. Internal pull-up and pull-down resistors disabled.

[8] 5 V tolerant pad with TTL levels and hysteresis and internal pull-up resistor.

[9] 5 V tolerant pad with 20 ns glitch filter providing digital I/O function with TTL levels and hysteresis.

[10] Pad provides special analog functionality. A 32 kHz crystal oscillator must be used with the RTC.

[11] When the system oscillator is not used, connect XTAL1 and XTAL2 as follows: XTAL1 can be left floating or can be grounded (grounding is preferred to reduce susceptibility to noise). XTAL2 should be left floating.

[12] When the RTC is not used, connect VBAT to $V_{DD(REG)(3V3)}$ and leave RTCX1 floating.